KERALA POLYTECHNIC SYLLABUS • REV2021

Code: 2091 **Semester:** 2 **Credits:** 3

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Course Overview

This course provides a foundational understanding of organic chemistry and polymer science, essential for Diploma students in Polymer Technology. It bridges the gap between basic chemical principles and the complex world of macromolecules.

Category	Engineering Science
Teaching Scheme	3 Hours / Week (L:3, T:0, P:0)
Total Periods	45 Periods per Semester
Prerequisites	Applied Chemistry (Classification of Polymers)

Course Outcomes (CO)

CO No.	Description	Duration	Cognitive Level
CO1	Explain the basic characteristics of carbon and organic compounds	10 Hours	Understanding
CO2	Describe the history, general characteristic and classification of polymers	10 Hours	Understanding
CO3	Explain the chemistry of polymerization and polymerization techniques	13 Hours	Understanding
CO4	Interpret the process of polymer degradation and its impact on environment	10 Hours	Applying

MODULE 1

Organic Chemistry & Petrochemicals

1.1 The Uniqueness of Carbon

Carbon is the backbone of organic chemistry due to two unique properties:

- **Tetravalency:** Carbon has 4 valence electrons, allowing it to form 4 covalent bonds.
- **Catenation:** The unique ability of carbon atoms to form long chains or rings by bonding with other carbon atoms.

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1.2 Classification of Organic Compounds

Type	Description	Examples
Alkanes	Saturated hydrocarbons (Single bonds)	Ethane (C_2H_6)
Alkenes	Unsaturated (Double bond)	Ethene (C_2H_4)
Alkynes	Unsaturated (Triple bond)	Ethyne (C_2H_2)
Aromatic	Ring structures with resonance	Benzene, Toluene

1.3 Petroleum and Petrochemicals

Petroleum is a complex mixture of hydrocarbons. Refining involves **Fractional Distillation** to separate components based on boiling points.

- **Cracking:** Process of breaking down heavy hydrocarbon molecules into lighter ones (e.g., producing Petrol from heavy oil).
- **Important Petrochemicals:** Ethylene, Propylene, Benzene, Styrene, and Phenol. These serve as raw materials for the polymer industry.

MODULE 2

History and Classification of Polymers

2.1 Basic Definitions

- **Monomer:** The small repeating unit that forms a polymer.
- **Polymer:** A high molecular weight macromolecule formed by the linkage of monomers.

- **Oligomer:** A low molecular weight polymer (usually 2-10 units).

2.2 Functionality

Functionality is the number of bonding sites available in a monomer.

- **Bifunctional:** Forms linear chains (e.g., Ethylene).
- **Polyfunctional:** Forms branched or cross-linked networks.

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2.3 Classification of Polymers

Basis	Types
Origin	Natural (Rubber, Cellulose), Synthetic (PVC, PE), Semi-synthetic
Thermal Behavior	Thermoplastic (Recyclable), Thermoset (Permanent set)
Structure	Linear, Branched, Cross-linked
Composition	Homopolymer (One type of monomer), Copolymer (Two or more)

Formula Bank & Calculations

Degree of Polymerization (DP)

$$DP = M / m$$

Where:
M = Molecular weight of Polymer
m = Molecular weight of Monomer

Worked Example:

Calculate the Degree of Polymerization (DP) for Polyethylene having a molecular weight of 28,000 g/mol. (Atomic weight: C=12, H=1)

Step 1: Find monomer weight (Ethylene: C_2H_4)
m = (2 × 12) + (4 × 1) = 24 + 4 = 28 g/mol.

Step 2: Apply formula
DP = 28,000 / 28 = 1,000.

Answer: The DP is 1000.

Visual Library

Polymer Architectures

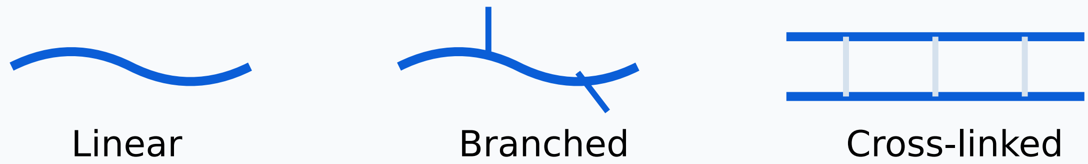

Figure 1: Comparison of Polymer Chain Arrangements.

MODULE 3

Polymerization Chemistry & Techniques

3.1 Types of Polymerization

- **Chain (Addition) Polymerization:** Monomers add one by one to a growing active center. No byproduct.
- **Step (Condensation) Polymerization:** Reaction between functional groups. Usually produces a small byproduct like water.

3.2 Free Radical Mechanism

1. **Initiation:** Formation of free radicals using initiators (e.g., Benzoyl Peroxide).
2. **Propagation:** Rapid growth of the chain.
3. **Termination:** Chain growth stops by coupling or disproportionation.

3.3 Polymerization Techniques

Technique	Description	Pros/Cons
Bulk	Pure monomer + initiator	High purity, but difficult heat removal.
Solution	Monomer dissolved in solvent	Easy heat control, but solvent removal is costly.
Suspension	Monomer droplets in water	Produces beads, easy to handle.

Emulsion	Uses soap (surfactant)	Fast reaction, high molecular weight.

MODULE 4

Polymer Degradation

Degradation is the deterioration of polymer properties due to environmental factors.

4.1 Types of Degradation

- **Thermal Degradation:** Breakdown due to heat.
- **Oxidative Degradation:** Reaction with atmospheric oxygen.
- **Photo Degradation:** Damage caused by UV rays from sunlight.
- **Hydrolytic Degradation:** Breakdown by water/moisture.

4.2 Anti-degradants

Chemicals added to polymers to prevent degradation:

- **Antioxidants:** Prevent oxidation.
- **Antiozonants:** Protect against ozone cracking (important for rubber).
- **UV Stabilizers:** Protect against sunlight.

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Module-wise Practice Questions

▶ **Q1. Explain Catenation in Carbon. (3 Marks)**

▶ **Q2. Distinguish between Thermoplastics and Thermosetting plastics. (5 Marks)**

▶ **Q3. What are the three steps in Free Radical Polymerization? (3 Marks)**

Practice Model Paper

(Based on REV2021 Pattern)

Part A (Short Answer - 3 Marks each)

1. Define Functionality with an example.
2. What is a Homopolymer? Give one example.
3. List two applications of Styrene.

Part B (Medium Answer - 5 Marks each)

1. Explain the fractional distillation of petroleum.
2. Describe the suspension polymerization technique.
3. Explain the environmental impact of polymer degradation.

Part C (Long Answer - 10 Marks each)

1. Classify polymers based on their structure and thermal behavior with suitable examples.
2. Explain the mechanism of free radical polymerization in detail.

Quick Revision

- **Carbon:** Tetravalent and shows catenation.
- **DP:** Number of monomer units in a chain.
- **Cracking:** Heavy oil → Light oil (Petrol).
- **Addition Polymer:** No byproduct (e.g., PE, PP).
- **Condensation Polymer:** Byproduct like H_2O (e.g., Nylon, PET).
- **Antioxidants:** Protect polymers from oxygen damage.

Glossary

Macromolecule
A very large molecule, such as a polymer, consisting of many smaller structural units.

Isomerism
Compounds having the same molecular formula but different structural arrangements.

Initiator
A chemical species that starts the polymerization reaction.

Official Source Declaration

This content is developed based on the official **SITTTR Kerala Revision 2021** syllabus for the course **Fundamentals Of Polymer Science (2091)**. All modules, outcomes, and technical topics align with the prescribed curriculum for the Diploma in Polymer Technology.

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- **[illegible] 3:** [illegible] [illegible] [illegible] [illegible] [illegible] (Polymerization Techniques).
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